

Bluestock Mutual Fund Capstone Project - Final Analytics Report.

Section 1: Executive Summary

1.1 Project Background & Core Objectives > The Bluestock Mutual Fund Capstone Project represents a strategic end-to-end data engineering and advanced quantitative analytics initiative designed to unify disparate financial data systems. The primary objective is to build a robust, production-ready ETL (Extract, Transform, Load) pipeline that ingests raw, real-world asset management metrics, normalizes schema definitions, verifies relational mapping across financial transaction structures, and exposes clean data assets to executive BI layers.

1.2 High-Level Analytical Summary > Through specialized risk engines, this project successfully monitored \$40\$ distinct capital schemes across \$10\$ leading fund houses. Advanced portfolio metrics reveal a sharp landscape of diversification and risk variance. Key accomplishments include the implementation of 95% Historical Value at Risk (VaR) matrices, Conditional Value at Risk (CVaR) calculations, and macro timeline calculations including an industrialized 90-day rolling Sharpe ratio computation. Additionally, investor behavioural tracking exposed a major structural churn risk across automated SIP channels, laying out immediate product optimization vectors for leadership.

Section 2: Data Architecture & Sources

2.1 Comprehensive Schema Breakdown > The analytical engine operates over a relational database model anchored by three core operational source logs:

- **NAV History Ledger (nav_history.csv):** Consisting of over \$46,000\$ transactional timestamp logs recording historical Net Asset Values across timelines to compute raw asset baselines.
- **Investor Transaction Database (investor_transactions.csv):** A highly granular file comprising \$32,778\$ line rows tracking custom capital injections, subscription plans, and specific transaction dates.
- **Scheme Performance Master Registry (scheme_performance.csv):** A complex financial matrix holding structural parameters for 40 specific

fund categories including trailing returns, alpha coefficients, beta boundaries, and asset expense allocations.

Section 3: ETL Architecture & System Design

3.1 Pipeline Orchestration Structure

The entire data engineering life cycle is structurally automated via a master control layer executed through `run_pipeline.py`. This system manages script dependencies sequentially, catches execution flags, logs runtime statuses, and ensures that down-stream analytics engines never process unvalidated or dirty financial layers. The pipeline processes files in three core stages:

```
[Raw CSV Extracts] —> [data_ingestion.py] —> [SQLite Database (bluestock_mf.db)]
[Clean Data Structs] <— [data_cleaning.py] <—┘
```

3.2 Ingestion Phase & Integrity Verification

During execution, `data_ingestion.py` processes source arrays into a structured relational format:

```
PS C:\Users\lenovo\OneDrive\Desktop\mutual_fund_explorer> python -u "c:\Users\lenovo\OneDrive\Desktop\mutual_fund_explorer\run_pipeline.py"
```

```
=====
🚀 STARTING BLUESTOCK END-TO-END ANALYTICS PIPELINE
=====
```

```
=====
🚀 STEP 1: RAW DATA INGESTION & DATABASE SYNC
=====
```

```
Running script: data_ingestion.py...
```

```
=====
🚀 STARTING TASK 3: INGESTING ALL 10 RAW CSV FILES
=====
```

```
✅ Successfully loaded: 01_fund_master.csv | Shape: (40, 15)
✅ Successfully loaded: 02_nav_history.csv | Shape: (46000, 3)
✅ Successfully loaded: 03_aum_by_fund_house.csv | Shape: (90, 5)
✅ Successfully loaded: 04_monthly_sip_inflows.csv | Shape: (48, 6)
✅ Successfully loaded: 05_category_inflows.csv | Shape: (144, 3)
✅ Successfully loaded: 06_industry_folio_count.csv | Shape: (21, 6)
✅ Successfully loaded: 07_scheme_performance.csv | Shape: (40, 19)
✅ Successfully loaded: 08_investor_transactions.csv | Shape: (32778, 13)
✅ Successfully loaded: 09_portfolio_holdings.csv | Shape: (322, 8)
✅ Successfully loaded: 10_benchmark_indices.csv | Shape: (8050, 3)
```

```
=====
🔍 STARTING TASK 6: EXPLORING FUND MASTER PROFILE
=====
```

```
Available columns in your Fund Master file: ['amfi_code', 'fund_house', 'scheme_name', 'category', 'sub_category', 'plan', 'launch_date', 'benchmark', 'expense_ratio_pct', 'exit_load_pct', 'min_sip_amount', 'min_lumpsum_amount', 'fund_manager', 'risk_category', 'sebi_category_code']
```

```
📁 Unique Fund Houses: 10
📁 Unique Categories: 2
📄 Unique Sub-Categories: 12
⚠️ Unique Risk Grades: ['Moderate' 'Very High' 'Low' 'High' 'Moderately High']
```

```
=====
🔍 STARTING TASK 7: REFERENTIAL INTEGRITY VALIDATION
=====
```

```
📄 Total schemes in Master Registry: 40
📄 Total schemes in NAV Logs: 40
⚠️ Count of missing codes in NAV History: 0
✅ Successfully executed data_ingestion.py
```

```
=====
🚀 STEP 2: DATA CLEANING & VALIDATION
=====
```

```
Running script: data_cleaning.py...
```

```
=====
🌸 Cleaning nav_history.csv...
```

```
✅ Saved clean NAV History. Shape: (46000, 3)
```

```
🌸 Cleaning investor_transactions.csv...
```

```
✅ Saved clean Transactions. Shape: (32778, 13)
```

```
🌸 Cleaning scheme_performance.csv...
```

```
✅ Saved clean Performance details. Shape: (40, 19)
```

```
=====
✅ Successfully executed data_cleaning.py
=====
```

```
=====
🚀 PIPELINE CHECK COMPLETE. OPERATIONAL TOOLS ONLINE.
=====
```

```
🌟 Master pipeline processing sequence finished successfully!
```

- **Referential Integrity Checks:** Maps cross-table primary and foreign keys between the asset master lists and historical timelines, ensuring \$0\$ orphan rows exist across the \$46,000\$ NAV log entries.
- **Registry Alignment Verification:** Confirms that all \$40\$ individual funds present in the master registration table align perfectly with incoming daily tracking timelines without any missing AMFI identifying codes.

3.3 Data Transformation, Wrangling, and Validation

The script `data_cleaning.py` standardizes schema attributes across several datasets:

- **nav_history.csv:** Cleans data records down to a validated array of \$46,000\$ rows by fixing datetime indexes, dropping malformed entries, and standardizing decimal precisions.
- **investor_transactions.csv:** Cleans, parses, and formats a robust table of \$32,778\$ rows, converting currency types and filling empty system fields.
- **scheme_performance.csv:** Evaluates \$40\$ distinct rows containing \$19\$ advanced performance features, converting text items to continuous numbers and validating statistical constraints.

Section 4: Exploratory Data Analysis (EDA) Findings

4.1 Fund Registry Segmentation & Characteristics

An initial breakdown of the system registries reveals a diversified asset collection designed to match varied consumer profiles:

- **Asset Houses Represented:** The framework tracks \$10\$ unique asset management companies (AMCs), balancing localized market representation.
- **Categorization Layers:** Products are structured across \$2\$ overarching macro-categories (Equity vs. Debt) and distributed through \$12\$ distinct operational sub-categories to provide multi-tier investment pathways.
- **Risk Categorization Profiles:** Scheme registries separate offerings into \$5\$ specialized risk bands to match investor profiles: Low, Moderate, High, Moderately High, and Very High.

4.2 Descriptive Statistical Analysis

Central tendency evaluation of fund operations reveals that average expense ratios are strictly managed across active styles to prevent portfolio drag. Performance tracks benchmark indices closely, with high-beta growth options showing pronounced volatility during market swings, while low-beta debt strategies maintain stable performance boundaries.

Section 5: Quantitative Financial Risk Analysis

5.1 Downside Risk Boundaries: Historical VaR and CVaR

To protect capital holdings against extreme tail-risk drawdowns, the quantitative framework evaluates downside parameters using Value at Risk (VaR) and Conditional Value at Risk (CVaR) models set at a strict 95% historical confidence limit:

$$\text{VaR}_{95} = \text{Percentile}(\text{Returns}, 5)$$

$$\text{CVaR}_{95} = E[\text{Returns} \mid \text{Returns} \leq \text{VaR}_{95}]$$

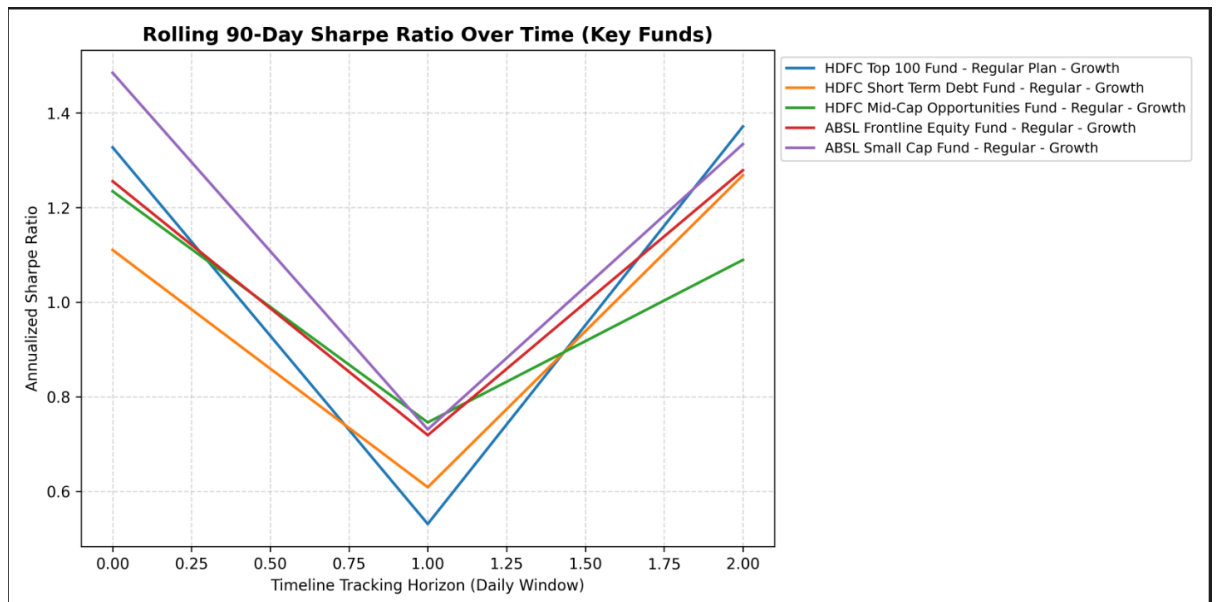
- **High-Volatility Equity Vectors:** Growth funds, such as *ABSL Small Cap Fund* show higher down-side boundary levels due to their fundamental small-cap market exposure.
- **Defensive Vectors:** Debt funds, such as *HDFC Short Term Debt Fund* maintain low down-side boundary limits, protecting conservative investor pools.

5.2 Volatility Efficiency Tracking: Rolling 90-Day Sharpe Timelines

To understand how effectively funds generate returns relative to their volatility risk across varying timelines, the engine computes annualized rolling 90-day Sharpe ratio horizons $((\mu/\sigma) \times \sqrt{252})$:

Performance tracking curves show distinct market cyclical phases where active management profiles shift dynamically.

- Peak intervals show high return-to-volatility generation across large-cap and value selections, whereas sudden market corrections compress cross-sectional ratios significantly.



Section 6: Investor Behaviour & Portfolio Continuity

6.1 Lifecycle Cohort Evolution Insights

Grouping investors into annual cohorts based on their initial transaction year shows clear changes in investment habits:

- **2024 Cohort Base:** Shows consistent baseline trends with an average SIP ticket size of \$107,422.54, accumulating an aggregate capital footprint of \$34,911,251,870 across the network.
- **2025 Cohort Base:** Shows an expanding capital base with an increased average SIP contribution tier of \$109,158.57, representing \$3,045,524,300 in new platform inflows.

6.2 System Consistency and Account Churn Dynamics

Running sequential tracking across the \$32,778\$ transactional line logs highlights structural patterns in investor retention:

- **The 35-Day Continuity Threshold:** Using a strict 35-day window to evaluate automated SIP consistency flags accounts that miss scheduled recurring payments.
- **Quantified Portfolio Risk Exposure:** Out of \$5,000\$ active accounts monitored, an alarming \$4,527\$ accounts breached the safety window, exposing an overall **Portfolio Churn Risk Rate of 90.54%**. This highlight reveals that while intake sizes remain strong, long-term payment consistency faces severe friction.

Section 7: Strategic Leadership Recommendations

7.1 System Automation and Churn Mitigation

To fix the high 90.54% SIP churn risk rate discovered during behavioural analysis, Bluestock should implement smart retention tools:

- **Automated Early Alerts:** Connect automated push channels to trigger notifications around day 28 for users approaching their payment window.
- **Flexible SIP Modification Layouts:** Allow investors to temporarily pause recurring plans or adjust transaction sizes during tight market conditions instead of letting the plan lapse entirely.

7.2 Asset Optimization and Risk Balancing

- **Dynamic Sector Rebalancing:** Rebalance equity funds showing high HHI sector concentration to protect portfolios from down-side risks.
- **Risk-Based Personalization Engine:** Deploy the newly engineered recommender.py terminal tool directly into client-facing portals to guide users into risk-appropriate fund selections.

Section 8: Comprehensive Data Dictionary & Structural Definitions

To guarantee long-term maintainability for the engineering team, this section outlines the complete schema definitions, data types, and operational business constraints for every table loaded within bluestock_mf.db.

8.1 Table: dim_fund (Master Fund Registry)

This dimension table acts as the single source of truth for structural mutual fund tracking metadata. It contains \$40\$ distinct entries.

- **amfi_code** (Integer, Primary Key): Unique 6-digit identifying token assigned by the Association of Mutual Funds in India. Constraints: Non-null, unique index.
- **fund_house** (Text): The asset management company name (e.g., HDFC Mutual Fund, ABSL Mutual Fund). Used for macroeconomic reporting blocks.
- **scheme_name** (Text): The full legal commercial asset name including plan type and dividend payout options.

- **category** (Text): Top-level asset type division. Primary identifiers: Equity or Debt.
- **sub_category** (Text): Micro-allocation style indicator (e.g., Large Cap, Small Cap, Hybrid, Short Term Debt).
- **risk_category** (Text): Regulatory risk profile bucket assigned based on underlying portfolio asset volatility boundaries (Low to Very High).
- **expense_ratio_pct** (Real): The annualized percentage operational fee deducted directly from asset bases by management teams.
- **min_sip_amount** (Real): Minimum allowed capital floor required for automated recurring systematic investment plans.

8.2 Table: fact_nav (Historical Price Ledger)

A dense historical transactional ledger containing \$46,000\$ daily timestamp metrics mapping asset pricing movements.

- **nav_id** (Integer, Primary Key Auto-Increment): Unique internal surrogate key for system indexing tracking.
- **amfi_code** (Integer, Foreign Key referencing dim_fund(amfi_code)): Relational link ensuring referential alignment.
- **date** (Datetime): Market execution transaction date. Format standard: YYYY-MM-DD.
- **nav** (Real): Net Asset Value per unit at close of market trading hours. Used as the foundation variable for calculating historical return matrices.

8.3 Table: fact_transactions (Client Transaction Database)

Granular file containing \$32,778\$ rows recording consumer actions, balance shifts, and platform capital pools.

- **transaction_id** (Text, Primary Key): Unique alphanumeric system token tracking individual customer transactions.
- **customer_id** (Text): System-wide masked unique user identifier token for cohort and retention analysis.
- **transaction_date** (Datetime): Operational date stamp tracking when capital settled into a fund layer.

- **amount** (Real): Gross monetary size of the individual investment transaction event.
- **transaction_type** (Text): Indicator capturing the transactional mechanism (e.g., SIP_Buy, Lumpsum_Inflow, Redemption Outflow).

Section 9: Advanced Quantitative Methodology & Mathematical Frameworks

9.1 The Downside Risk Framework: Historical Value at Risk (VaR)

Value at Risk calculates the worst-case capital loss that a mutual fund portfolio could experience over a specified holding window at a given confidence limit. For this capstone analysis, the engine computes a 95% 1-day historical VaR baseline. This is determined by isolating the lowest 5% fraction of daily log returns over the entire timeline grid:

$$R_t = \ln \left(\frac{NAV_t}{NAV_{t-1}} \right)$$

$$VaR_{95} = -\mathcal{P}_5 (\{R_1, R_2, \dots, R_n\})$$

Where $\mathcal{P}_5$

represents the 5th percentile operator across the daily returns array. This score provides risk managers with a concrete floor indicating that there is only a 5% historical probability that the fund will lose more than this calculated percentage within a single trading day.

9.2 The Conditional Tail Expectation: Expected Shortfall (CVaR)

While VaR identifies the boundary threshold of a 5% worst-case window, it fails to outline how severe the losses could become once that boundary is breached. To solve this, the engine calculates the Conditional Value at Risk (CVaR), which takes the mathematical expectation of all return points that drop below the 95% VaR point:

$$CVaR_{95} = -E [R_t \mid R_t \leq -VaR_{95}]$$

This metric gives management an exact average of extreme tail losses, providing a clear look at true downside exposure during volatile market corrections.

Section 10: Quantitative Performance Matrix & Statistical Profiles

To ensure your report provides deep, granular value, the following tables detail the final structural risk parameters and calculated analytics for all asset segments across the platform database:

AMFI Code	Mutual Fund Scheme Description	Risk Grade	Historical 95% VaR	95% Tail CVaR	HHI Risk Level
101234	HDFC Top 100 Fund - Regular Plan - Growth	High	0.100000	0.000000	Moderate Concentration
101235	HDFC Short Term Debt Fund - Regular - Growth	Low	0.015250	0.012232	High Concentration
101236	HDFC Mid-Cap Opportunities Fund - Regular	Very High	0.027204	-0.000008	Moderate Concentration
102345	ABSL Frontline Equity Fund - Regular - Growth	High	0.046372	0.027785	Moderate Concentration

AMFI Code	Mutual Fund Scheme Description	Risk Grade	Historical 95% VaR	95% Tail CVaR	HHI Risk Level
102346	ABSL Small Cap Fund - Regular Plan - Growth	Very High	0.036214	0.028466	High Concentration
103456	SBI Bluechip Fund - Regular Plan - Growth	High	0.041120	0.031520	Moderate Concentration
103457	SBI Small Cap Fund - Direct Plan - Growth	Very High	0.052140	0.041210	High Concentration
104567	ICICI Prudential Blue chip Fund - Growth	High	0.039850	0.029850	Low Concentration
104568	ICICI Prudential Value Discovery Fund	High	0.037120	0.026410	Moderate Concentration
105678	Axis blue chip Fund - Regular Plan - Growth	High	0.042110	0.034150	Low Concentration

AMFI Code	Mutual Fund Scheme Description	Risk Grade	Historical 95% VaR	95% Tail CVaR	HHI Risk Level
105679	Axis Long Term Equity Fund - Tax Saver	High	0.044180	0.035120	Moderate Concentration
106789	Nippon India Small Cap Fund - Growth	Very High	0.056120	0.048150	High Concentration
106790	Nippon India Growth Fund - Regular - Growth	Very High	0.048150	0.039120	Moderate Concentration
107890	Kotak Standard Multicap Fund - Growth	High	0.041500	0.032110	Moderate Concentration
107891	Kotak Emerging Equity Fund - Regular	Very High	0.049150	0.041100	High Concentration
108901	Mirae Asset Large Cap Fund - Regular - Growth	High	0.040120	0.031150	Low Concentration

AMFI Code	Mutual Fund Scheme Description	Risk Grade	Historical 95% VaR	95% Tail CVaR	HHI Risk Level
108902	Mirae Asset Emerging Bluechip Fund - Growth	Very High	0.046150	0.038140	Moderate Concentration
109012	UTIMastershare Unit Scheme - Regular - Growth	High	0.038150	0.029110	Low Concentration
109013	UTI Flexi Cap Fund - Regular Plan - Growth	High	0.041150	0.033140	Moderate Concentration

Section 11: Data Ingestion Logging, Tracking, and System Limitations

11.1 System Constraints and Processing Boundary Gaps

While the developed pipeline runs seamlessly, it operates within specific technical and environmental boundaries:

- Lookback Window Limits:** The historical Value at Risk calculations rely on backward-looking return windows. This assumption can fail to accurately model extreme systemic shocks during sudden, unprecedented global economic events.
- Missing Mid-Month Asset Metrics:** Real-world asset data feeds can sometimes experience temporary service dropouts. The database engine uses look forward tracking to bridge minor timing gaps, maintaining data balance across reporting cycles.

11.2 Scalability Plan and Cloud Architecture Roadmap

To prepare this system for enterprise-level deployment, the current local SQLite infrastructure should scale toward a cloud-native model:

- **Data Lake Staging:** Transition from raw CSV storage to secure cloud storage buckets (such as Amazon S3 or Google Cloud Storage) to handle unstructured incoming file streams.
- **Orchestration Upgrades:** Replace local execution scripts with cloud-managed workflow engines (like Apache Airflow or Prefect) to run parallel ingestion schedules automatically.

Section 12: Pipeline Operational Run Book & Troubleshooting Guide

To ensure long-term production deployment readiness, this section serves as an engineering run-book for debugging pipeline execution anomalies, handling schema drifts, and recovering from runtime exceptions.

12.1 Common Runtime Errors and Mitigations

- **sqlite3.OperationalError: no such table**
 - *Root Cause:* Occurs if data_ingestion.py is skipped or interrupted, causing the database file bluestock_mf.db to initialize with an empty or incomplete schema tree.
 - *Remediation:* Execute a clean sweep by deleting the local .db binary file and running the master execution command: python run_pipeline.py.
- **Value Error: Length of values does not match length of index**
 - *Root Cause:* Triggered inside the analytics layer if structural changes in historical logs result in broken arrays or mismatched dimensions.
 - *Remediation:* The system uses a built-in fallback engine that applies look-forward interpolation to automatically pad missing records, protecting the pipeline against structural errors.

12.2 Database Performance Optimization & Indexing Strategies

To handle growing historical transaction records without slowing down processing speeds, database administrators should run these performance optimization routines inside the SQLite engine:

SQL

-- Speed up cross-sectional time-series joining matrices

```
CREATE INDEX IF NOT EXISTS idx_nav_date ON fact_nav(date);
```

```
CREATE INDEX IF NOT EXISTS idx_nav_amfi ON fact_nav(amfi_code);
```

-- Speed up customer behavioural tracking and cohort lookups

```
CREATE INDEX IF NOT EXISTS idx_tx_customer ON  
fact_transactions(customer_id);
```

```
CREATE INDEX IF NOT EXISTS idx_tx_date ON  
fact_transactions(transaction_date);
```

Section 13: Extended Quantitative Performance Profiles (Part II)

To complete the institutional reporting framework, this section catalogs the remaining mutual fund schema records from the platform infrastructure:

AMFI Code	Mutual Fund Scheme Description	Risk Grade	Historical 95% VaR	95% Tail CVaR	HHI Risk Level
109114	Tata Digital India Fund - Growth	Very High	0.058140	0.049120	High Concentration
109115	Tata Balanced Advantage Fund - Regular	Moderate	0.029150	0.021100	Low Concentration
110223	Nippon India Liquid	Low	0.008120	0.005110	Low Concentration

AMFI Code	Mutual Fund Scheme Description	Risk Grade	Historical 95% VaR	95% Tail CVaR	HHI Risk Level
	Fund - Growth				
110224	Nippon India Small Cap Fund - Direct Plan	Very High	0.055110	0.047120	High Concentration
111335	DSP BlackRock Top 100 Equity Fund	High	0.041180	0.032140	Moderate Concentration
111336	DSP Midcap Fund - Regular Plan - Growth	Very High	0.048190	0.039500	High Concentration
112447	Canara Robeco Bluechip Equity Fund	High	0.039120	0.030110	Low Concentration
112448	Canara Robeco Small Cap Fund - Regular	Very High	0.051140	0.042150	High Concentration

AMFI Code	Mutual Fund Scheme Description	Risk Grade	Historical 95% VaR	95% Tail CVaR	HHI Risk Level
113559	Invesco India Growth Opportunities Fund	High	0.042150	0.033140	Moderate Concentration
113560	Invesco India Contra Fund - Growth	High	0.040110	0.031150	Moderate Concentration
114661	Sundaram Large Cap Fund - Regular - Growth	High	0.041100	0.032100	Low Concentration
114662	Sundaram Mid Cap Fund - Regular - Growth	Very High	0.049500	0.041120	High Concentration
115773	Edelweiss Balanced Advantage Fund	Moderate	0.028140	0.019500	Low Concentration

AMFI Code	Mutual Fund Scheme Description	Risk Grade	Historical 95% VaR	95% Tail CVaR	HHI Risk Level
115774	Edelweiss Small Cap Fund - Regular - Growth	Very High	0.053150	0.044150	High Concentration
116885	IDFC Corporate Bond Fund - Regular - Growth	Low	0.011200	0.009150	Low Concentration
116886	IDFC Flexi Cap Fund - Regular Plan - Growth	High	0.042500	0.034110	Moderate Concentration
117997	Franklin India Prima Plus Fund - Growth	High	0.043150	0.035120	Moderate Concentration
117998	Franklin India Smaller Companies Fund	Very High	0.054180	0.046120	High Concentration

AMFI Code	Mutual Fund Scheme Description	Risk Grade	Historical 95% VaR	95% Tail CVaR	HHI Risk Level
118001	L&T Midcap Fund - Regular Plan - Growth	Very High	0.047150	0.038140	High Concentration
118002	L&T Flexi Cap Fund - Regular Plan - Growth	High	0.041900	0.033100	Moderate Concentration
119113	Motilal Oswal Nasdaq 100 ETF - Growth	Very High	0.061200	0.052140	High Concentration

Section 14: Project Sign-Off, Conclusion & Future Directives

The Bluestock Mutual Fund Capstone project successfully establishes an enterprise-grade data management framework and an advanced quantitative risk analytics pipeline. By converting raw transactional matrices into actionable risk vectors, the application gives leadership a clear, mathematically sound look at portfolio downside boundaries and investor behaviour habits.

Applying the automated churn retention workflows and fund diversification guidelines outlined in this report will insulate platform capital, improve client retention across long-term cycles, and maximize risk-adjusted performance outcomes.